

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

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Query 1: -CREATE DATABASE

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree with categories like MANAGEMENT, INSTANCE, and PERFORMANCE. The center pane shows a SQL script with two lines: `CREATE DATABASE bookflow4_db;` and `USE bookflow4_db;`. The right pane shows a message: 'Automatic context help is disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.' Below the script, the 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	09:34:10	CREATE DATABASE bookflow4_db	1 row(s) affected	0.047 sec
2	09:34:10	USE bookflow4_db	0 row(s) affected	0.000 sec

Query 2: Create Books Table (DDL + Constraints)

The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the 'Server Enterprise' tree. The center pane shows a SQL script with the following lines: `CREATE TABLE BOK (`, `book_id INT PRIMARY KEY,`, `title VARCHAR(100) NOT NULL,`, `isbn VARCHAR(20) UNIQUE,`, `published_year INT CHECK (published_year < 2027)`, and `);`. The right pane shows a message: 'Automatic context help is disabled. Use the toolbar manually get help for the current caret position or toggle automatic help.' Below the script, the 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	09:34:10	CREATE TABLE BOK	1 row(s) affected	0.047 sec

Output: -

✓	4	09:38:44	CREATE TABLE Bok (book_id INT PRIMARY KEY...	0 row(s) affected	0.109 sec
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Query 3: Insert Books and display

```
11 ✖ 2520030071
12 INSERT INTO Bok (book_id,title,isbn,published_year) VALUES
13 (1,'The Great Bay','9780743273565',2000),
14 (2,'The Indian Culture','2345987456',2010),
15 (3,'1947','9456783457934',2020);
16 • SELECT * FROM Bok;
```

Result Grid	Filter Rows:	Edit:	Export/Import:
book_id	title	isbn	published_year
1	The Great Bay	9780743273565	2000
2	The Indian Culture	2345987456	2010
3	1947	9456783457934	2020
*	NULL	NULL	NULL

Result Grid

Form Editor

Field Types

Output:-

✓	12	12:10:49	INSERT INTO Bok (book_id,title,isbn,published_y...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.062 sec
✓	13	12:10:49	SELECT * FROM Bok LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Query 4: Members Table: Create

```
1 CREATE TABLE Members(
2 member_id INT PRIMARY KEY,
3 full_name VARCHAR(100),
4 email VARCHAR(100) UNIQUE
5 );
```

Output:-

✓	19	12:20:46	CREATE TABLE Members(member_id INT PRIM...	0 row(s) affected	0.063 sec
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Query 5:- Insert Members table

```
INSERT INTO Members (member_id, full_name, email) VALUES
(101, 'G. Hemanth Venkatesh', 'hemanthvenkatesh5554@gmail.com'),
(102, 'Rohit Verma', 'rohitverma@gmail.com'),
(103, 'Vivek Sharma', 'viveksharma@gmail.com');
```

Output: -

✓	19	12:20:46	INSERT INTO Members (member_id,full_name,email) VALUES ...	3 row(s) affected	Records: 3	Duplicates: 0	Warnings: 0	0.016 sec
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Query 6: - : Display Members table

```
SELECT * FROM Members;
```

Output: -

	member_id	full_name	email
▶	101	G. Hemanth Venkatesh	hemanthvenkatesh5554@gmail.com
	102	Rohit Verma	rohitverma@gmail.com
	103	Vivek Sharma	viveksharma@gmail.com
*	NULL	NULL	NULL

Members 3 × Apply Revert Context Help Snippets

Output: -

✓	20	12:22:31	INSERT INTO Members (member_id,full_name,email) VALUES ...	3 row(s) affected	Records: 3	Duplicates: 0	Warnings: 0	0.016 sec
✓	21	12:24:30	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned				0.000 sec / 0.000 sec

Query 7: Create Loans Table

```
1 CREATE TABLE Loans (
2     loan_id INT PRIMARY KEY,
3     member_id INT NOT NULL,
4     amount DECIMAL(10,2) NOT NULL CHECK (amount > 0),
5     loan_date DATE NOT NULL,
6     FOREIGN KEY (member_id) REFERENCES Members(member_id)
7 );
```

Output: -

✓	22	12:26:05	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_...	0 row(s) affected				0.078 sec
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```
CREATE TABLE Loan (  
    loan_id INT PRIMARY KEY,  
    member_id INT,  
    book_id INT,  
    loan_date DATE,  
    FOREIGN KEY (member_id) REFERENCES Members(member_id),  
    FOREIGN KEY(book_id) REFERENCES Bok(book_id)  
);
```

Output:-

23 12:33:08 CREATE TABLE Loan (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN... 0 row(s) affected

Query 8: Insert 10 Loans

```
INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
(1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
(3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),  
(5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),  
(7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),  
(9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

Output: -

91 13:41:27 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, ... 10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0 0.04 sec

Query 9: Display

```
SELECT * FROM Loans;
```

Output :-

Result Grid	Filter Rows:	Edit:	Export/Import:
loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	103	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01
NULL	NULL	NULL	NULL

Loans 7 x

Apply Revert

Query 10: JOIN Query

```
45  SELECT m.full_name AS Member_Name, b.title AS Book_Title
46  FROM loans l
47  INNER JOIN Members m ON l.member_id = m.member_id
48  INNER JOIN Bok b ON l.book_id = b.book_id;
```

Output: -

Result Grid				Filter Rows:	Export:	Wrap Cell Content:	Result Grid Form Editor Field Types
	Member_Name	Book_Title					
	Adama Samarth	The Great Bay					
	Adama Samarth	The Indian Culture					
	Adama Samarth	1947					
	Adama Samarth	The Great Bay					
	Rohit Verma	The Indian Culture					
	Rohit Verma	The Great Bay					
	Rohit Verma	1947					
	Vivek Sharma	1947					
	Vivek Sharma	The Indian Culture					
	Vivek Sharma	The Great Bay					
Result 14							Read Only
✓	103	13:50:44	SELECT m.full_name AS Member_Name,b.title AS Book_Title FROM loans l INNER JOIN Members m ON l.member_id = m.member_id # INNER JOIN Bok b ON l.book_id = b.book_id			10 row(s) returned	

Query 11: GROUP BY Query

```
1  SELECT published_year, COUNT(book_id) AS Total_Books
2  FROM Bok
3  GROUP BY published_year
4  ORDER BY published_year;
```

Output: -

Result Grid				Filter Rows:	Export:	Wrap Cell Content:	Result Grid Form Editor
	published_year	Total_Books					
	2000	1					
	2010	1					
	2020	2					
Result 3							Read Only
✓	104	13:51:12	SELECT published_year, COUNT(book_id) AS Total_Books FROM Bok GROUP BY published_year ORDER BY published_year			3 row(s) returned	